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Econometrics Course

Are there differences in the Drivers of CDS Returns? Evidences from US and EU Markets

Group 4

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Introduction and Literature Review

A credit default swap (CDS) transfers the default risk of a reference entity. Its spread is the periodic premium paid for protection and therefore provides a market-based measure of perceived credit risk. This project studies the daily five-year CDS spread returns of large US and European banks from January 2015 to September 2021. The research questions are:

which common financial factors explain daily bank CDS returns, and can those factors forecast the next-day return of a regional CDS index?

The analysis is motivated by structural credit-risk theory. In the framework of [5], stronger equity values reduce default risk, whereas asset volatility increases it. Empirical studies show that equity returns, volatility and interest rates explain an important part of credit-spread and CDS-spread changes [2, 3]. Focusing specifically on banks, [1] find that both market-wide conditions and bank-specific factors determine euro-area bank CDS spreads. More recent international evidence confirms the importance of market conditions and firm performance, especially during crisis periods [4].

Following this literature, the project combines regional equity returns, the VIX, gold returns and yield-curve factors. It contributes a direct US–Europe comparison and evaluates several methods: panel regression, residual and ARCH diagnostics, GARCH-X, random forest and one-day-ahead AR(1)-GARCH-X forecasting.

Data and Methodology

Dataset and Sample Construction

The main dataset consists of daily CDS quotes across ten maturities (from 1 to 10 years) for 661 firms operating in a wide range of sectors, spanning the period from January 2015 to September 2021. We decided to focus our analysis on the five-year CDS spreads, denoted by $PX5_{i,t}$, for five US and seven European banks. The five-year maturity is generally among the most liquid CDS contracts and provides a balance between short-term and long-term credit risk expectations, avoiding maturities that are either too close to expiration or too distant in time. The US sample includes: JPMorgan Chase, Bank of America, Citigroup, Goldman Sachs and Wells Fargo. The European sample includes: Banco Santander, BNP Paribas, Deutsche Bank, Intesa Sanpaolo, UniCredit, BBVA and Barclays.

In addition to CDS spreads, we collect a set of financial variables associated with changes in perceived credit risk. The table below summarizes the data sources, transformations, and economic interpretation of each variable used throughout the empirical analysis.

Table 1.1: Data sources and transformations

Variable	Source	Transformation	Economic role
Five-year CDS	Provided CDS dataset	$r_{i,t}^{CDS} = (PX5_{i,t})$	Change in perceived bank credit risk
S&P 500, EUROSTOXX	Yahoo Finance	Daily log return	Regional equity-market conditions
VIX	Daily VIX dataset	First difference	Change in expected market uncertainty
Gold	Intraday gold dataset	Daily close and log return	Global risk and safe-haven factor
US rates	US Treasury dataset	Standardization and PCA	US yield-curve level and slope
EU rates	Custom ECB dataset	Standardization and PCA	European yield-curve level and rotations

CDS levels are sorted by bank and date before calculating log-differences. The other variables contain observations for the same dates as the CDS.

Panel and Aggregate Series

Panel regressions use CDS returns separated by banks. Models requiring a single dependent series instead use an equally weighted CDS index:

$$CDS_t^{EW} = \frac{1}{N}\sum_{i=1}^{N} PX5_{i,t}, \qquad r_t^{EW} = (CDS_t^{EW}).$$

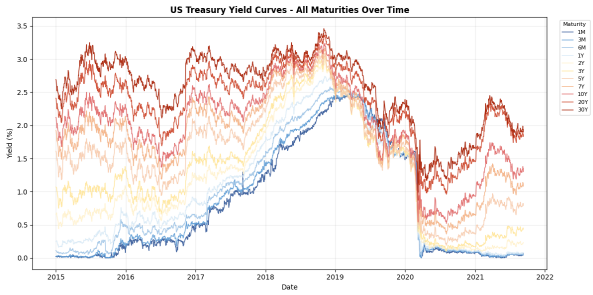
Equal weighting gives the same values for the banks instead of accounting for capitalization.

Yield-Curve Dimension Reduction

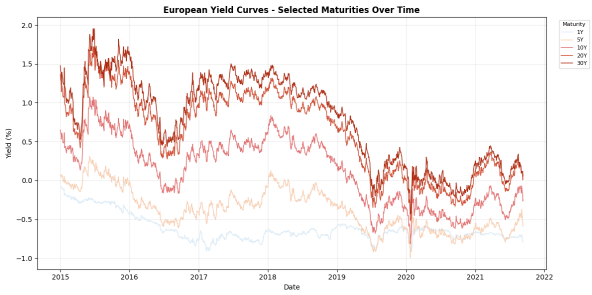
The US curve contains eleven maturities from one month to thirty years. The European curve contains one-, five-, ten-, twenty- and thirty-year rates. Including all maturities would create strong multicollinearity, because yields at nearby maturities move together. Principal component analysis (PCA) is therefore applied to each standardized yield curve. If Z_t is the vector of standardized yields, the k-th component is

$$PC_{k,t} = w_k' Z_t,$$

where w_k maximizes explained variance subject to orthogonality with the previous components.



(a) US Treasury yields.



(b) European yields.

The first two PCs retain 99.09% of US yield variation and 98.67% of European yield variation. PC1 is almost perfectly correlated with the average yield-curve level in both regions (0.9984 in the US and 0.9968 in Europe). US PC2 is strongly correlated with a long-short slope proxy (0.9252). European PC2 has a lower slope correlation (0.5190), so it also captures more complex curve rotations.

Econometric and Machine-Learning Methods

Panel regression. As a starting point, we model bank CDS returns as a function of several market-wide factors that capture overall financial conditions:

$$r_{i,t}^{CDS} = \alpha_i + \beta_1 PC1_t + \beta_2 PC2_t + \beta_3 r_t^{Equity} + \beta_4 \Delta VIX_t + \beta_5 r_t^{Gold} + \varepsilon_{i,t}.$$

Both fixed-effects (FE) and random-effects (RE) estimators are fitted with standard errors. The Hausman test evaluates whether the RE estimator is consistent. Residual adequacy is examined using a Breusch–Pagan-style heteroskedasticity diagnostic, Durbin–Watson statistic and Jarque–Bera normality test.

GARCH. The aggregate CDS return displays volatility clustering. After an ARCH-LM test, its conditional mean and variance are modeled as

$$\begin{aligned} r_t^{EW} &= \mu + \gamma' X_t + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \quad z_t \sim N(0,1), \\ \sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \end{aligned}$$

The explanatory GARCH model includes contemporaneous VIX, equity and gold variables. The forecasting version additionally uses an AR(1) term and all predictors lagged by one day, including the two rate PCs.

Random forest. A random forest regressor captures nonlinearities and interactions without imposing a functional form. The model uses 1,000 trees, maximum depth six and at least ten observations per terminal leaf. Performance is summarized by in-sample R^2 and correlation. Impurity-based and permutation importance are compared because the former can favor variables that permit many possible splits.

Forecast evaluation. The AR(1)-GARCH model is estimated using the first 90% of observations and evaluated on the final 10%. The metrics are

$$\text{MAE} = \frac{1}{T}\sum_t |y_t - \hat{y}_t|, \qquad \text{RMSE} = \sqrt{\frac{1}{T}\sum_t (y_t - \hat{y}_t)^2},$$

alongside out-of-sample R^2, forecast-realization correlation and directional accuracy. Separating the final chronological block is essential because random train-test splitting would leak future information into a time-series model.

We also evaluate the volatility forecasts generated by the GARCH component. Since daily conditional variance is not directly observable, one-day-ahead conditional variance forecasts are compared with squared realized CDS-index returns, which are interpreted as a noisy proxy for realized variance.

Expected Signs

We expect equity returns to have a negative effect on CDS spreads, as stronger stock-market performance is generally associated with lower perceived credit risk: increasing equity value moves the firm farther from default. An increase in the VIX should widen CDS spreads by signaling higher uncertainty. Gold and yield-curve effects may represent several channels, including safe-haven demand, monetary policy and bank profitability.

Results

Panel Regression and Hypothesis Testing

Table 1.2: Fixed-effects panel estimates

	United States		Europe	
Regressor	Coefficient	p-value	Coefficient	p-value
Rates PC1	0.0012	0.2317	0.0098	0.0019
Rates PC2	-0.0019	0.1592	-0.0150	0.0002
Equity return	-0.5657	0.0000	-0.5973	0.0000
VIX change	0.0446	0.0000	0.0490	0.0000
Gold return	0.0638	0.0000	-0.0119	0.0394
Within R^2	0.3673		0.3880	
Observations	8,730		12,184	

Equity returns and VIX changes are the most stable drivers. A one-unit increase in the regional equity log return is associated with approximately a 0.57 to 0.60 reduction in the CDS log return, while an increase in the VIX is associated with wider CDS spreads. Both effects are highly statistically significant and have the signs predicted by credit-risk theory.

Rate factors differ across regions. Neither US rate PC is significant at the 5% level, whereas both European PCs are significant. A higher European level factor is associated with wider CDS spreads and a higher PC2 with narrower spreads. Gold is significant in both regressions but changes sign: positive in the US and slightly negative in Europe. These contrasting effects suggest that gold and rates capture region-dependent channels rather than a single universal credit-risk factor.

FE and RE estimates are practically identical. The Hausman statistics equal 0.0000 for the US and 0.0049 for Europe, with p-values of 1.0000. Thus, the null that RE is consistent is not rejected. Poolability-test p-values are also approximately one, suggesting that the included common market variables explain most of the systematic daily variation captured by the model.

Table 1.3: Residual and ARCH diagnostics

Statistic	United States	Europe
Durbin–Watson	1.9515	1.7572
Jarque–Bera	27,184.22	243,843.29
Residual kurtosis	11.60	24.91
ARCH-LM TR^2 (10 lags)	491.73	268.14
ARCH-LM p-value	< 0.001	< 0.001

Normality is strongly rejected and residuals are heavy-tailed, especially in Europe. The

ARCH-LM tests reject constant conditional variance in both regional indices, motivating the GARCH specifications.

GARCH Results

Table 1.4: Contemporaneous GARCH models for equally weighted CDS indices

Statistic	United States	Europe
Mean-equation R^2	0.405	0.471
VIX-change coefficient	0.0646	0.0727
Equity-return coefficient	-152.4679^{***}	-169.0431^{***}
Gold-return coefficient	25.0372^{***}	-8.7276
ω	0.2296^{***}	0.6362^{*}
α	0.1080^{***}	0.1287^{***}
β	0.8497^{***}	0.7643^{***}
$\alpha + \beta$	0.9577	0.8930

Notes: *, ** and *** denote statistical significance at the 10%, 5% and 1% levels, respectively.

The regional equity return remains the dominant mean-equation variable after conditional heteroskedasticity is modeled. Its negative coefficient is highly significant in both regions. Gold remains important only in the US model, while the VIX loses significance. This suggests that some explanatory power previously attributed to the VIX reflected changing volatility rather than only movements in the conditional mean.

The variance parameters are economically important. Both α and β are positive and significant, confirming that volatility reacts to new shocks and remains persistent over time. The sums $\alpha + \beta = 0.9577$ in the US and 0.8930 in Europe are below one, so conditional variance is mean reverting, but the decay of shocks is slow. Persistence is particularly high for the US CDS index.

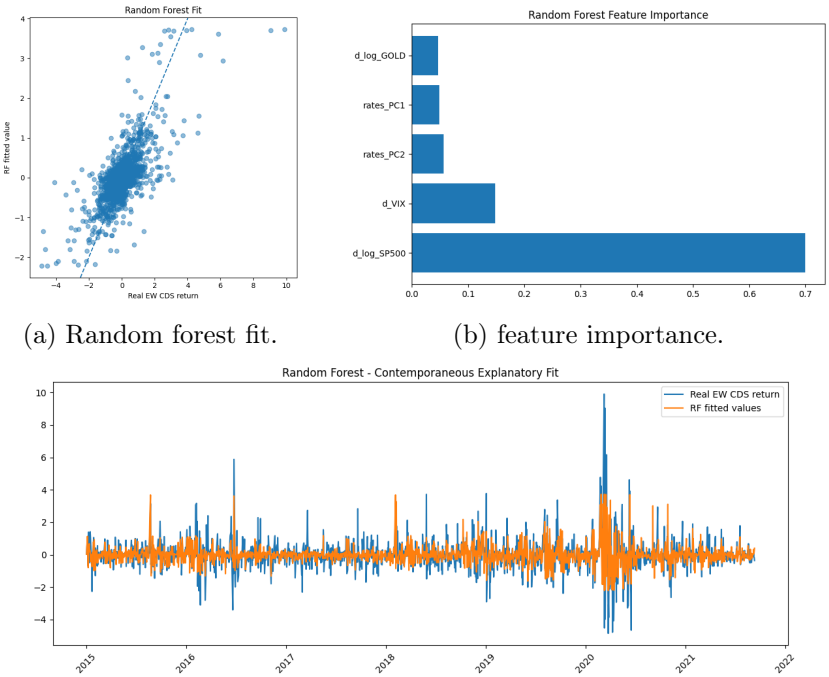
The mean-equation R^2 rises from 0.3673 in the US panel model to 0.405 in the aggregate US GARCH model and from 0.3880 to 0.471 in Europe. These figures are not perfectly comparable because the dependent series and estimation methods differ, but they indicate that modeling the regional index and its conditional variance captures additional structure.

Interpretation

Taken together, the panel and GARCH results agree with the literature: equity-market performance is closely connected to credit-risk repricing [3, 5], and volatility is persistent. The difference between US and European rate coefficients may reflect different monetary environments and bank business models. However, the regressions are contemporaneous; coefficients describe conditional associations and should not be interpreted as causal effects.

Random Forest Results

The random forest is used as a flexible explanatory benchmark on the equally weighted CDS return. It reaches an in-sample R^2 of 0.5157 and a fitted-realized correlation of 0.7229 in the US. In Europe, the corresponding values are 0.5306 and 0.7303. The nonlinear model therefore fits the observed data better than the linear specifications, although its in-sample performance cannot be interpreted as evidence of forecasting ability.



(a) Random forest fit. (b) feature importance.

Figure 1.2: US random forest fit and feature importance.

The importance rankings reinforce the econometric findings. For the US, the S&P 500 return accounts for 0.6996 of impurity-based importance and 0.6775 of permutation importance. The VIX ranks second; rate PCs and gold have smaller contributions. The European result is even more concentrated: EUROSTOXX reaches 0.8876 impurity importance and 0.9428 permutation importance.

Agreement between impurity and permutation measures makes the ranking more credible. More importantly, agreement between the random forest and panel regression indicates that the central role of equity markets is not an artifact of an imposed linear functional form. Nonetheless, correlated predictors can divide importance between themselves, and the selected hyperparameters were not evaluated through time-series cross-validation.

One-Day-Ahead Forecasting Results

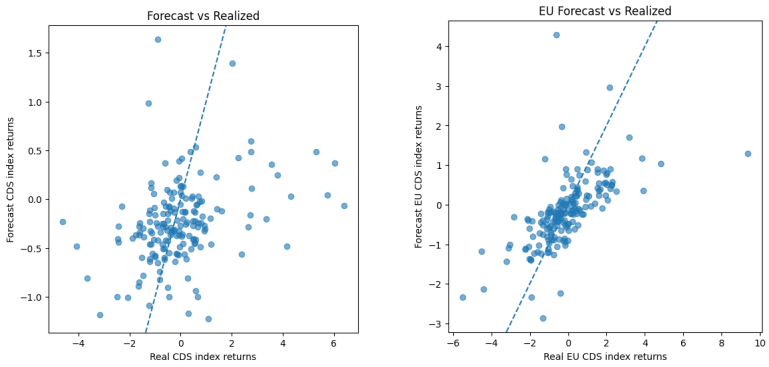
For forecasting, the contemporaneous model is replaced by an AR(1)-GARCH model using one-day-lagged explanatory variables:

$$r_t^{EW} = c + \phi r_{t-1}^{EW} + \gamma' X_{t-1} + \varepsilon_t, \qquad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

The final 10% of observations are held out and never used for estimation.

Table 1.5: AR(1)-GARCH out-of-sample performance

Metric	United States	Europe
MAE	1.0293	0.8512
MSE	2.3723	1.6380
RMSE	1.5402	1.2798
R^2	0.0918	0.3865
Correlation	0.3450	0.6336
Directional accuracy	62.86%	81.71%



(a) US forecast versus realized CDS-index returns.

(b) EU forecast versus realized CDS-index returns.

Figure 1.3: AR(1)-GARCH one-day-ahead forecasts.

The US model has limited ability to predict return magnitude, with an out-of-sample R^2 of 0.0918 and correlation of 0.3450. It nevertheless predicts the sign correctly in 62.86% of cases. The European model performs substantially better on every metric: its R^2 is 0.3865, correlation is 0.6336 and directional accuracy reaches 81.71%. In its estimated mean equation, the lagged CDS return, lagged VIX change and lagged EUROSTOXX return are statistically significant.

Table 1.6: AR(1)-GARCH-X volatility forecast performance

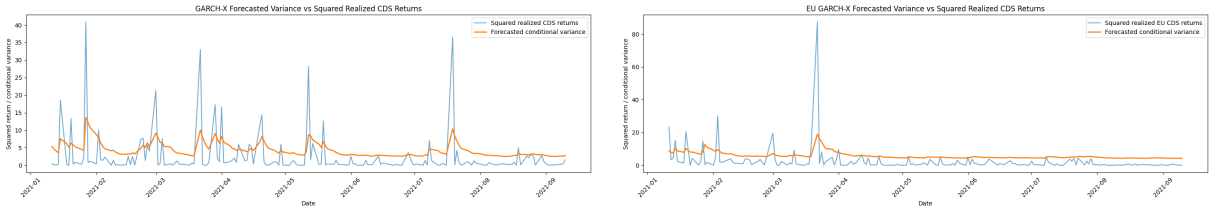
Metric	United States	Europe
Correlation: forecast variance vs. r_t^2	0.6433	0.6007
MSE variance forecast	28.7342	53.1811
RMSE variance forecast	5.3604	7.2925
GARCH persistence: $\alpha + \beta$	0.9428	0.9071

The volatility-forecast results are more mixed than the return-forecast results. The correlation between forecasted conditional variance and squared realized returns is positive in both regions, equal to 0.6433 for the United States and 0.6007 for Europe. This suggests that the AR(1)-GARCH models capture part of the broad volatility clustering in CDS-index returns.

However, the graphical comparison shows that the forecasted variance is considerably smoother than realized squared returns. Squared returns display sharp isolated spikes,

while model-implied variance reacts more gradually and remains elevated after large shocks. This behaviour is consistent with the high estimated GARCH persistence, with $\alpha + \beta$ equal to 0.9428 in the US model and 0.9071 in the European model.

Overall, the volatility forecasts are useful for identifying periods of higher conditional risk, but they should not be interpreted as precise predictors of daily volatility spikes. The models tend to smooth extreme movements and may keep predicted volatility high for longer than suggested by squared realized returns.



(a) US forecasted conditional variance versus squared realized CDS-index returns.

(b) EU forecasted conditional variance versus squared realized CDS-index returns.

Figure 1.4: AR(1)-GARCH one-day-ahead conditional variance forecasts.

Discussion, Robustness and Limitations

Three broad conclusions emerge from the results. First, regional equity returns are the most consistent explanatory driver of bank CDS returns. Their negative coefficient is economically coherent and robust across panel regression, GARCH and random forest. Second, volatility clustering is substantial: constant-variance regression is inadequate, and GARCH persistence is high in both regions. Third, predicting the returns is basically worthless, as we should expect in an efficient market. However the model seemed capable to predict the sign better than random.

Main limitations and robustness checks.

- The sample includes the COVID-19 crisis. We decided to include the first COVID wave to have a better forecast for the second wave.
- GARCH innovations are assumed Gaussian, so we decided to use a robust Covariance estimator. We tried also a t-Student model but with a not noticeable effect. Other models considering fatter tails could be implemented
- Volatility forecasts are evaluated against squared daily returns, which are a noisy proxy for realized variance.
- The equally weighted index ignores differences in bank size and CDS market liquidity. Market-value or notional weighting could alter results.
- Forecasting performance should also be compared with explicit benchmarks such as zero return, historical mean and a simple AR(1).
- Common regressors vary only over time, while standard errors are clustered by bank. Alternative two-way clustering or time clustering would better address cross-sectional dependence generated by common shocks.

These limitations qualify the strength of the conclusions but do not overturn the central descriptive finding.

Conclusion

This project examined the determinants and one-day-ahead predictability of five-year CDS returns for major US and European banks between 2015 and 2021. The dataset combined bank CDS spreads with regional equity indices, the VIX, gold and yield curves. PCA reduced the highly correlated rate maturities to two factors while preserving more than 98% of yield-curve variation.

The explanatory analysis provides a consistent answer to the first research question. Stronger regional equity returns are associated with narrower CDS spreads, while increases in the VIX are associated with spread widening. Rate factors are significant in Europe but not in the US. Residual diagnostics reject normality and constant conditional variance, and GARCH estimates show high volatility persistence. Random forests improve in-sample fit but confirm, rather than overturn, the dominant role of equity markets.

The answer to the forecasting question is more nuanced. The US AR(1)-GARCH model has modest predictive power in term of returns, confirming the hypothesis of an efficient CDS market. The predictions about volatility seems to be completely out of range in term of magnitude but capable to capture the sign better than random.

Overall, the results support the economic framework suggested by credit-risk theory and previous empirical CDS research. Future work should focus on crisis-specific structural breaks, heavy-tailed GARCH innovations, alternative index weights and stricter rolling-window forecast comparisons.

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